Native ordered prime-norm radix-14 evaluator: report

Question and scope

Can an ordered fourteen-position prime-norm scalar format be evaluated natively using three shared digit atlases for radices `394-tau`, `380-tau` (twice), and `371-tau` (eleven times)? The overlay is hash-guarded to its parent source. It adds position-specific windows, atlas lookup, and a bounded terminal lookup; the original fixed-format evaluator remains the comparison reference.

Correctness and shape checks

The native implementation passed exhaustive residue/congruence/norm checks for atlas norms 154,057, 143,263 and 136,531, plus an independent 128-point binary scalar holdout. The panel verified all 12,288 outputs in every arm. The parent experiment's exact shape screen and bound Lean theorem are linked from the README; they establish the candidate residue/radius facts but do not formalize the concrete native selector and curve implementation. A candidate-specific Lean refinement remains open.

Direct point payload is 22,267,136 bytes, 46,784 bytes less than fixed radix-14's 22,313,920 bytes. Total retained native memory, including three digit atlases and terminal metadata, is 26,556,708 bytes, 520,720 bytes more than the fixed format's 26,035,988.

Qualified result

The frozen same-point Tier 1 panel completed 96 valid arms and 48 pairs. Every adjacent A/A control passed `[0.90, 1.11]`; source, binary and fixture artifacts stayed stable.

Metric	Ordered candidate	Fixed reference
Mean online time, 12,288 scalars	137.806005 ms	135.634388 ms
Mean time per scalar	11.214681 µs	11.037955 µs
Mean preparation	3,971.307 ms	3,956.667 ms
Retained bytes	26,556,708	26,035,988

The paired candidate/reference throughput ratio is **0.985551×**, with repeat-bootstrap 95% interval [**0.979842, 0.988419**]. This is a qualified **throughput regression** of about 1.45% under the frozen hot-table workload and observed cloud-instance envelope. It is not an improvement over the fixed format. One-call latency and Tier 0 physical-host isolation were not measured. The committed analyzer regenerated `tier1-ordered-analysis.json` byte-for-byte and checks the panel outputs, hashes, pair order and calibration gates.

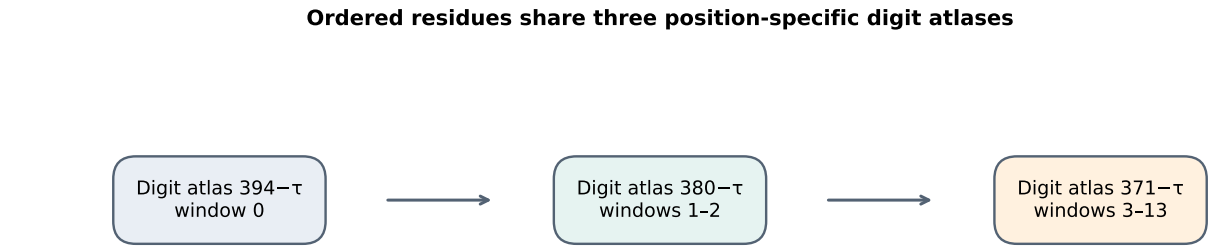
Graph and result index

The local graph shows the measured candidate/reference ratio and its interval below one; the atlas diagram describes which position windows share each radix. The canonical experiment index in `experiments/prime-j0-fourteen-position-mixed` now links this native evaluation as a separately scoped negative result. No prior graph source or end-to-end cost model is changed: the candidate's native hot-table comparison did not qualify as a pipeline improvement, and the README keeps the fixed format as the stronger measured option.

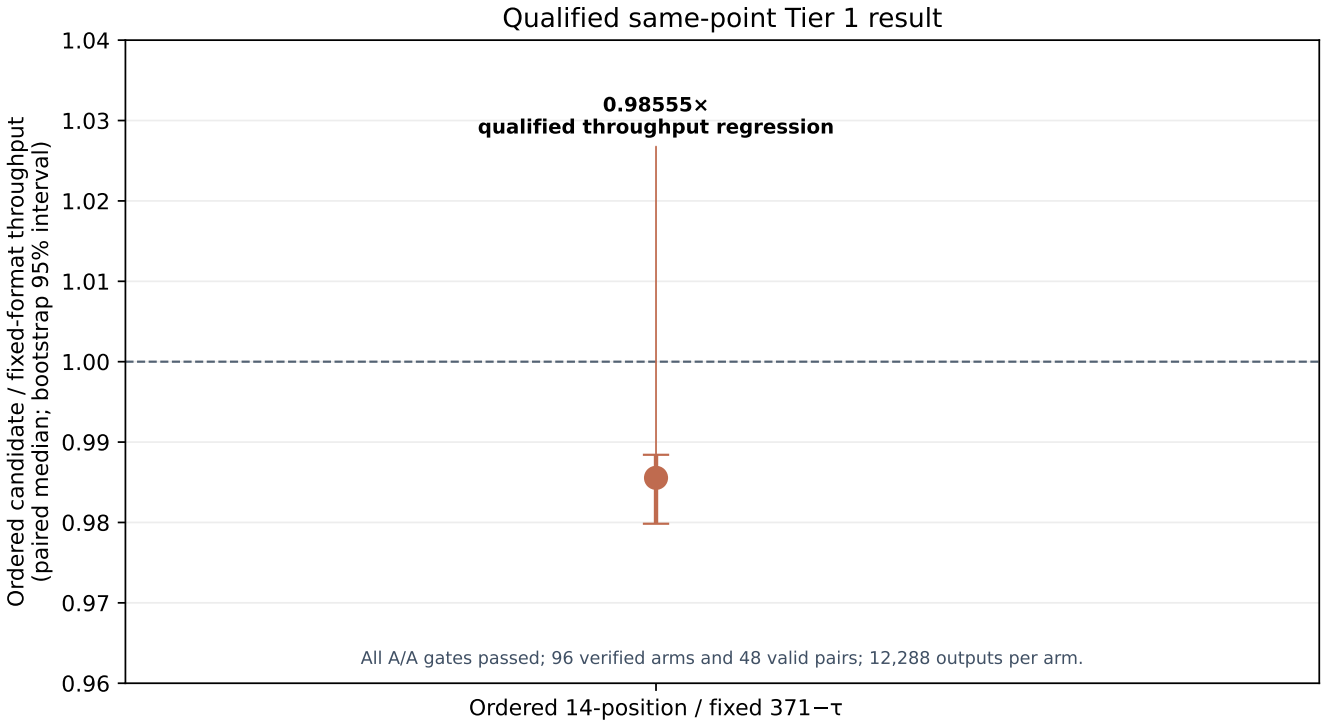
Rebuild with:

```
python3 experiments/prime-j0-ordered-native-20261015/visuals.py
pandoc experiments/prime-j0-ordered-native-20261015/REPORT-2026-10-19.md \
  --from=gfm --pdf-engine=xelatex -V geometry:margin=0.75in \
  -o experiments/prime-j0-ordered-native-20261015/REPORT-2026-10-19.pdf
```

Atlas diagram (SVG) · PDF



Measured throughput result (SVG) · PDF



Decision and next step

Retain the implementation, proofs, and negative qualified measurement as a record. The fixed `371-tau` format remains the stronger measured option under this panel's resource envelope. A future variant should first remove terminal metadata and atlas cost or reduce online work, then repeat the same-point calibrated comparison before reconsidering selection.